



# The constructive, overreliant, and irresponsible use of artificial intelligence tools in academia: Personality correlates and implications for academic integrity

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## ABSTRACT

The integration of AI into academia presents both opportunities and challenges for academic integrity. The literature review suggests that AI usage in academia may align with three general profiles—Constructive (e.g., using AI to enhance academic skills), Overreliant (e.g., heavy dependence on AI for academic tasks), and Irresponsible (e.g., using AI-generated content without verification)—each with significant implications for academic integrity. To expand our understanding of this subject, this study was conducted with two objectives: first, to introduce and validate the Academic AI Usage Inventory (AAUI), and second, to examine relationships between AI usage profiles and personality features, including maladaptive personality traits, psychopathic traits, attachment styles, and self-esteem. Data were collected online from 171 university students (aged 18–52;  $M = 29.19$ ,  $SD = 7.70$ ). Confirmatory factor analysis (CFA) supported the three-factor structure of the AAUI, which demonstrated acceptable internal consistency. Correlations and multiple regression analyses revealed that Disinhibition and Antagonism were positively linked to Irresponsible usage, with Disinhibition also associated with Overreliant usage. Conversely, Anankastia was positively related to Constructive usage and negatively to Irresponsible usage. Additionally, Callous-Unemotional (CU) and Impulsive-Irresponsible (II) traits were positively associated with Irresponsible usage, while II also correlated with Overreliant usage. Furthermore, Anxious attachment was associated with Overreliant and Irresponsible usage, while higher self-esteem predicted Constructive usage, and lower self-esteem was linked to Overreliant and Irresponsible profiles. Findings highlight the role of personality features in various AI engagement styles in academia, underscoring the need for policies to promote responsible AI usage and academic integrity.

## 1. Introduction

While preparing this work, AI tools were employed to assist with proofreading. Following their use, the author reviewed and revised the content as necessary, assuming full responsibility for the final version of the article. Prior to the release of OpenAI's ChatGPT in November 2022, such a statement might have seemed far-fetched—yet, in the wake of its introduction, the landscape of academic work has transformed in profound ways. This transformation has sparked intense discussions about the role of AI in academia, raising questions about its potential to revolutionize scholarly work and the ethical dilemmas it introduces (Cotton et al., 2024; Sullivan et al., 2023). Nevertheless, as a tool of immense power, AI can be seen as a double-edged sword in the context of academia (Shen et al., 2023). For instance, AI tools offer robust support for scholars in various academic tasks, such as drafting articles,

theses, or class projects, by assisting in tasks like brainstorming ideas, identifying and correcting grammatical errors, improving clarity, optimizing structure, and streamlining the proofreading process.

However, despite their impressive capabilities, AI tools also come with significant limitations. One major limitation is their tendency to produce seemingly credible but incorrect responses, such as inventing terms or information. This phenomenon, known as the "hallucination effect," is a common issue among many natural language processing models. Since these models rely on patterns in their training data without truly understanding the content, they may confidently fabricate information when faced with gaps in their knowledge (Rohrbach et al., 2018; Xiao & Wang, 2021). Relatedly, ChatGPT explicitly informs users that "ChatGPT can make mistakes. Check important info," emphasizing the need for critical review and verification of AI-generated content to ensure its accuracy and reliability. Furthermore, rather than engaging in

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genuine interaction, AI tools like ChatGPT primarily follow instructions and can sometimes make assumptions about user intentions instead of seeking clarification when provided with insufficient information. Without detailed prompts or clear constraints, these tools may deliver incomplete or inaccurate results (e.g., Shen et al., 2023). Moreover, the reliability of AI-generated responses is closely tied to the quality of the training data. If the data contains biases, inaccuracies, or misconceptions, these flaws can manifest in the model's outputs, underscoring the importance of scrutinizing both the technology and the information it provides (e.g., Shen et al., 2023). Importantly, while some AI tools have access to the Internet and can generate content with references, many of these advanced features require paid subscriptions. Unfortunately, these subscription-based services are often unaffordable for researchers in low-income countries, leaving them reliant on free versions of AI tools. These free versions, however, come with significant limitations. For instance, ChatGPT, which is trained on data available up to October 2023, cannot incorporate information from studies or developments published after that time.

In conclusion, the preceding discussion suggests that while AI tools offer significant advantages in academic work, their limitations remind us that AI cannot replace human judgment and expertise. Consequently, researchers must approach AI-generated content with a critical eye, carefully verifying the accuracy and ensuring that AI tools are used to complement, rather than replace rigorous academic processes. Ultimately, while AI can be a valuable asset in the academic realm, it is essential to maintain a balanced approach that integrates AI's capabilities with human insight and discernment.

In addition to these concerns, there is increasing apprehension that the growing reliance on AI tools for academic purposes may lead to excessive dependence, resulting in potential misuse and overuse (Kasneci et al., 2023; King & ChatGPT, 2023). This overreliance on AI has been linked to negative outcomes, including diminished critical thinking and problem-solving skills (Zhang et al., 2024). For example, in the study by Zhang et al. (2024) with 300 university students, participants highlighted several adverse effects of AI dependency, including laziness, decreased creativity, misinformation, weakened critical and independent thinking, lower engagement in information-seeking, plagiarism, potential copyright issues, difficulty in problem-solving, and reduced abilities in information judgment. Furthermore, it is reasonable to assume that scholars may become overly reliant on AI tools because of their ability to provide quick, direct answers, which can boost academic performance in the short term (Alshater, 2022; Quintans-Júnior et al., 2023). Additionally, AI tools are highly effective in tasks like reviewing, revising, grammar checking, and enhancing texts, often surpassing human abilities, especially for non-native English scholars aiming to submit manuscripts to English-language journals. In this scenario, these individuals might develop a sense of inferiority compared to AI, which may further entrench their dependence and create a cyclical pattern resembling addiction.

Building on the preceding discussion, academic AI usage could be categorized into three distinct profiles: Constructive, Overreliant, and Irresponsible.<sup>1</sup> Individuals with a constructive profile approach AI tools as valuable assistants, using them to enhance their existing skills without compromising their intellectual independence. They prioritize their own effort, employing AI for supplementary tasks like brainstorming or refining ideas, but maintain ownership of the final outcome. These users focus on personal growth and view AI as a secondary aid rather than a

primary driver of their academic work. This balanced and responsible approach ensures that AI supports their creativity rather than replacing it. Individuals with an Overreliant profile are heavily dependent on AI tools for academic tasks. Users in this category may feel that completing projects without AI is overly challenging or stressful, leading to a reliance on AI for generating ideas, structuring arguments, or producing significant portions of their work. Over time, this dependency can diminish confidence, motivation, and critical thinking abilities. Such reliance could risk creating a cycle where individuals view AI as essential, reducing their engagement with independent problem-solving and academic exploration. Finally, the users with the Irresponsible academic AI usage profile demonstrate a problematic and ethically questionable use of AI tools. They may rely on AI to complete substantial portions of their work without adequate review or verification, potentially introducing inaccuracies. These individuals often prefer to conceal their AI usage, even when transparency is expected, and may use AI-generated content, such as citations or references, without confirming its validity. This behavior reflects a disregard for ethical academic practices and a tendency to misuse AI for convenience, prioritizing quick results over accountability (see Fig. 1 for a visual representation of the three academic AI usage profiles).

Given the significance of these diverse academic AI usage profiles for both the academic community and individuals themselves, it becomes crucial to investigate the personality factors that may correlate with these patterns. This can offer valuable insights into the psychological mechanisms driving these behaviors. One set of factors that may contribute to understanding of different academic AI usage profiles involves maladaptive personality traits, as outlined in the Alternative Model of Personality Disorder (AMPD) in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association (APA); 2013)* and the International Classification of Diseases 11th Revision (ICD-11; World Health Organization (WHO); 2019) as part of their diagnostic approach to personality disorders. These traits are described as maladaptive variants of the Five Factor Model (FFM) personality traits and include Negative Affectivity (e.g., anxiousness, emotional instability; FFM variant of Neuroticism vs. Emotional Stability), Detachment (e.g., social withdrawal, anhedonia; FFM variant of Introversion vs. Extraversion), Antagonism (e.g., manipulateness, deceitfulness; FFM variant of Antagonism vs. Agreeableness), Disinhibition (e.g., impulsivity, distractibility; FFM variant of Disinhibition vs. Conscientiousness), Psychoticism (e.g., cognitive, perceptual dysregulation, eccentricity; FFM variant of Openness vs. Closedness), and Anankastia (e.g., rigidity, perfectionism; FFM variant of Conscientiousness).

Theoretically, maladaptive personality traits could exhibit varying relationships with the three academic AI usage profiles (i.e., Constructive, Overreliant, and Irresponsible). For instance, Negative Affectivity, characterized by heightened emotional instability and anxiousness, may align with the Overreliant profile. Individuals high in this trait might lean heavily on AI tools as a coping mechanism for academic stress or self-doubt. While this dependence may provide temporary relief, it could ultimately undermine their capacity for independent academic work. These theoretical expectations are consistent with empirical findings showing that elevated levels of neuroticism significantly increase the likelihood of using AI for academic tasks (Rodríguez-Ruiz, Espejo Siles, & Marín-López, 2024). In contrast, Detachment, marked by social withdrawal and disengagement, may correspond more closely to the Irresponsible profile, as such individuals might use AI as a substitute for collaboration and critical thinking, leading to less conscientious and ethical use of these tools. Nevertheless, empirical studies are needed to test these potential relationships.

Moreover, Antagonism, involving traits like manipulateness and deceitfulness, is particularly relevant to Irresponsible usage. Those high in Antagonism may misuse AI for unethical purposes, such as concealing their reliance on AI or engaging in plagiarism, thereby prioritizing self-interest over academic integrity. Supporting this perspective,

<sup>1</sup> It is important to note that the proposed profiles—Constructive, Overreliant, and Irresponsible—are not intended to establish rigid or mutually exclusive categories of academic AI usage. Instead, these profiles serve as a conceptual framework to guide exploration and hypothesis testing within this study. This approach is exploratory in nature and aims to provide initial insights into patterns of AI utilization, which can be further refined and validated in future research.

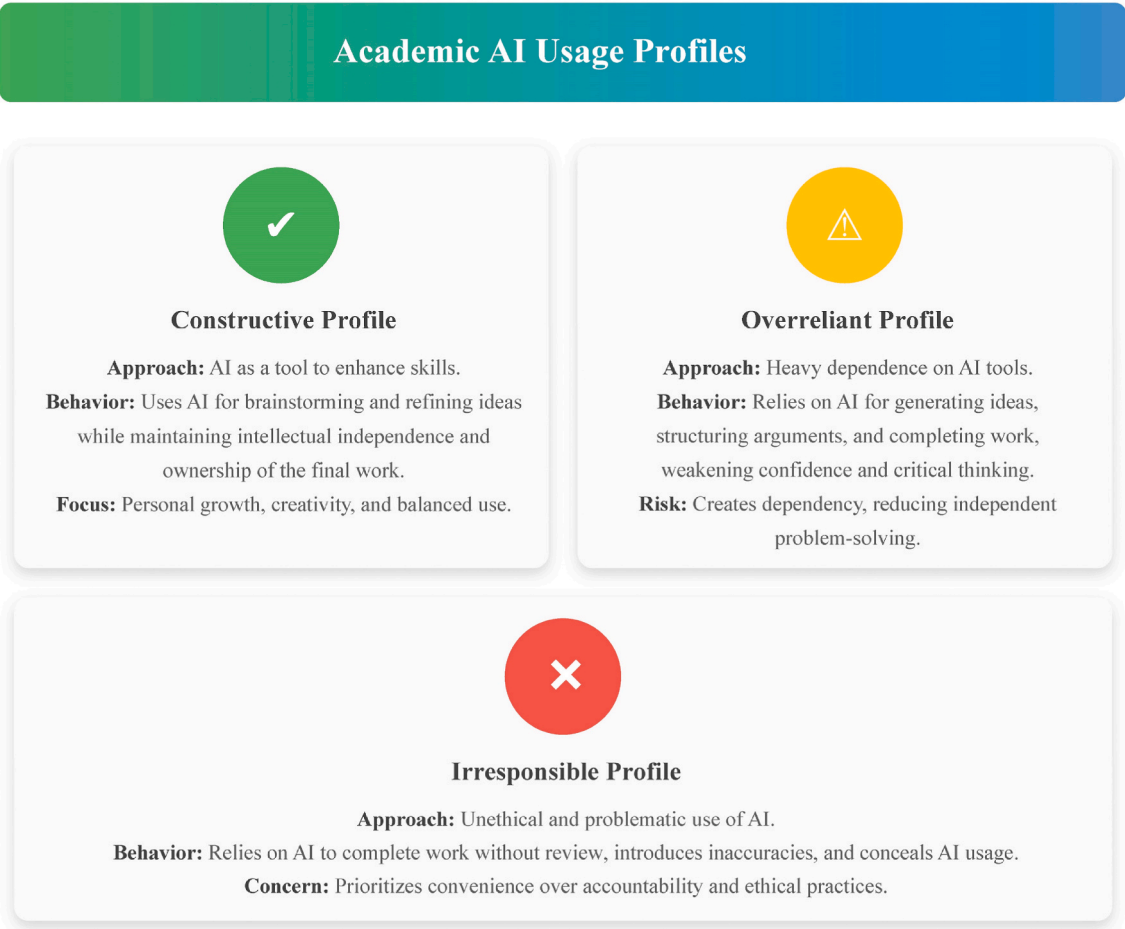


Fig. 1. Visual representation of the three academic AI usage profiles.

Greitemeyer and Kastenmüller (2023) found that traits associated with the Dark Triad—narcissism, Machiavellianism, and psychopathy—were positively linked to the intention to use chatbot-generated texts for academic misconduct. Likewise, de Winter et al. (2024) reported a moderate correlation between Machiavellianism and an opportunistic attitude towards the undisclosed use of ChatGPT.

Furthermore, Disinhibition, defined by impulsivity and lack of self-control, could relate to both Overreliant and Irresponsible usage. On the one hand, impulsive tendencies might drive unregulated or excessive reliance on AI. On the other hand, distractibility may hinder deliberate, constructive engagement with these tools. Consistent with this, Greitemeyer and Kastenmüller (2023) identified a negative relationship between conscientiousness—especially honesty-humility—and the misuse of AI for academic cheating. Similarly, Rodríguez-Ruiz, Espejo Siles, and Marín-López (2024) demonstrated that the use of AI to generate fake content was negatively associated with conscientiousness. Furthermore, Psychoticism, which encompasses cognitive and perceptual dysregulation, may also relate to Irresponsible usage. Individuals with high levels of this trait might develop unrealistic expectations of AI, uncritically accepting outputs that are flawed or unreliable. Conversely, Anankastia, characterized by perfectionism and rigidity, might support Constructive usage when paired with a disciplined approach to AI as a supplemental tool. Nevertheless, empirical studies are needed to test these theoretical hypotheses.

In addition to maladaptive personality traits, psychopathic personality traits could also play a significant role in shaping patterns of academic AI usage. Psychopathy is typically characterized by a combination of interpersonal (e.g., grandiosity and deceitfulness), affective (callousness, lack of guilt), behavioral (impulsivity and irresponsibility),

and antisocial behavior features (Hare, 2003). Theoretically, these traits may intersect with the Overreliant and Irresponsible profiles in distinct ways. For instance, interpersonal traits, such as manipulateness and deceitfulness, may align with the Irresponsible profile, as individuals high in these traits might misuse AI for unethical purposes, including fabricating work or concealing AI reliance. Also, affective traits, marked by a lack of guilt or concern for ethical norms, could similarly contribute to unethical AI usage, reflecting indifference to academic integrity. Likewise, behavioral traits like impulsivity and carelessness might relate to both Overreliant and Irresponsible profiles, driving excessive reliance on AI or careless and unethical application of its tools. Studies have identified a negative relationship between psychopathic personality traits and the misuse of AI for academic cheating (e.g., Greitemeyer & Kastenmüller, 2023). However, these studies have predominantly treated psychopathy as a unidimensional construct, neglecting its multidimensional nature (for a review see Elhami Athar, 2025) which warrants further exploration in future research.

In addition, attachment styles, which influence how individuals relate to others, could also play a role in shaping patterns of academic AI usage. Collins (1996) categorized attachment styles into three types: Close, Dependent, and Anxious, each of which may theoretically align with the profiles of academic AI usage in distinct ways. Individuals high in the close subscale, who feel comfortable with intimacy and collaboration, may be more likely to engage in Constructive AI usage. Their comfort with closeness may encourage them to seek support from peers or mentors, rather than excessively relying on AI tools. Conversely, those high in the anxious subscale, who experience fear of abandonment or failure, may be more susceptible to both Overreliant and Irresponsible AI usage. Their anxiety could drive them to use AI tools as a source of

constant reassurance, potentially leading to overuse or unethical practices such as academic dishonesty. The reliance on AI to alleviate academic stress or provide validation may hinder their development of critical thinking skills and ethical decision-making. Still, empirical studies are necessary to further explore and validate these potential relationships.

Finally, self-esteem—a psychological trait that shapes individuals' positive or negative evaluations of themselves (Rosenberg, 1965) — may also be an important factor in academic AI usage. Due to the emerging nature of this topic, limited research exists on whether self-esteem influences the motivation to utilize AI in academic settings. For instance, Bin-Nashwan et al. (2023) identified a significant positive relationship between self-esteem and ChatGPT usage among academics. Interestingly, their findings suggested that this relationship is moderated by academic integrity, indicating that higher levels of perceived integrity in using ChatGPT may boost self-esteem, thereby encouraging its adoption as an innovative tool. Nonetheless, empirical research is essential to examine how self-esteem interacts with academic AI usage, particularly in relation to the profiles discussed earlier.

### 1.1. The current study

To address the aforementioned gaps in the literature, this study pursued two primary objectives.

1. To develop and validate an *a priori* three-factor measure designed to assess the three patterns of academic AI usage—Constructive, Overreliant, and Irresponsible—using confirmatory factor analysis (CFA).
2. To explore the relationships between these AI usage profiles and key psychological variables, including maladaptive personality traits, psychopathic personality traits, attachment styles, and self-esteem, through Pearson correlation coefficients and multiple regression analyses.

Given the novelty of this research and the limited empirical evidence, hypotheses regarding the relationships between study variables were primarily derived from theoretical frameworks and the scarce existing empirical findings discussed above. Specifically, it was hypothesized that.

- Antagonism and Disinhibition would be positively associated with the Overreliant and Irresponsible AI usage profiles.
- Anankastia would be negatively associated with Overreliant and Irresponsible profiles but positively associated with Constructive usage.
- The behavioral dimension of psychopathy would relate positively to Overreliant usage, while the interpersonal and affective dimensions would be positively associated with Irresponsible usage.
- Due to limited prior findings, no specific hypotheses were made regarding attachment styles.
- High self-esteem would be positively associated with Constructive usage and negatively associated with the Overreliant profile.

## 2. Method

### 2.1. Participants and procedure

A total of 171 university students ( $M = 29.19$ ,  $SD = 7.70$ ; 64.33 % female) ranging in age from 18 to 52 years participated in this cross-sectional study. Based on sample size calculation guidelines, this sample size seems adequate for the statistical analyses conducted in this study, including factor analysis and regression analysis. Muthén and Muthén (2002) recommend a minimum of approximately 150 participants for a sufficient confirmatory factor analysis (CFA), which this study meets. Additionally, a commonly accepted guideline suggests

recruiting 5 to 15 participants per item or variable, further supporting the adequacy of the sample size for both CFA and regression analyses. Participants were recruited online using a convenience sampling method between October 2023 and August 2024. The invitation link for the study was distributed to university students in Iran via social media platforms, including Instagram and Telegram groups affiliated with universities. To be eligible, participants had to be enrolled at a university in Iran and have experience using AI tools for academic tasks, such as preparing manuscripts for journal submission, completing class projects, or working on thesis research.

Before participation, individuals were presented with a study description outlining the research objectives, inclusion criteria, and confidentiality assurances. Informed consent was obtained electronically, with participants confirming their voluntary participation by checking a box. The survey was structured in a way that required participants to answer all questions before submitting their responses, thereby ensuring that no data was missing. All procedures adhered to the ethical guidelines outlined in the World Medical Association's Declaration of Helsinki.

### 2.2. Measures

#### 2.2.1. Academic AI Utilization Inventory (AAUI)

The AAUI is a 17-item self-report measure specifically developed for this study to assess three distinct profiles of academic AI usage: Constructive, Overreliant, and Irresponsible. Items are rated on a 5-point Likert scale ranging from 1 ("Strongly Disagree") to 5 ("Strongly Agree") and load onto three subscales of Constructive (4 items), Overreliant (7 items), and Irresponsible (6 items).

The AAUI items were developed based on the definitions of the three profiles outlined in the Introduction section. The initial pool consisted of 17 items, which were reviewed by expert colleagues to ensure clarity and relevance. Feedback from these reviews resulted in refinements to enhance item comprehensibility. The finalized AAUI was then administered alongside other measures in this study and underwent psychometric evaluation. A confirmatory factor analysis (CFA) was conducted using JASP (version .19.2.0; JASP Team, 2024) to evaluate the fit of the *a priori* hierarchical three-factor model of the AAUI. The results indicated an acceptable fit, with a Comparative Fit Index (CFI) of .952, a Tucker-Lewis Index (TLI) of .944, and a Root Mean Square Error of Approximation (RMSEA) of .066 (Bentler, 1990; Hu & Bentler, 1999). Table 1 presents the AAUI items, along with the standardized item loadings, and Fig. 2 provides a visual representation of the AAUI's factor structure model. While most items surpassed the minimum loading threshold of .30 (Brown, 2015), items 1 and 13 fell slightly below this criterion. However, these two items were retained in the model due to their near-threshold loadings and the potential for further examination in future studies. The development process for the AAUI aligns with established guidelines for scale construction (e.g., Downing & Haladyna, 2006).

#### 2.2.2. The Personality Inventory for DSM-5 and ICD-11 – Brief Form Modified (PID5BF + M; Bach et al., 2020)

The PID5BF + M is a self-report measure consisting of 36 items designed to measure six broad domains of maladaptive personality traits, as specified in the DSM-5 AMPD and the ICD-11. These domains include Negative Affectivity (e.g., "I get emotional easily, often for very little reason"), Antagonism (e.g., "I'm good at conning people"), Psychoticism (e.g., "I have several habits that others find eccentric or strange"), Disinhibition (e.g., "I feel like I act totally on impulse"), Detachment (e.g., "I keep my distance from people"), and Anankastia (e.g., "It is important to me that things are done in a certain way"). Participants rate each item on a 4-point Likert scale, ranging from 0 ("very false or often false") to 3 ("very true or often true"), with higher scores reflecting a greater degree of personality pathology.

**Table 1**  
Standardized item loadings for the three-factor model of the AAUI (n = 171).

| Subscales     | Item Number | Item Content                                                                                                                                                                        | Loading |
|---------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Constructive  | 1           | I consider AI an assistive tool that can enhance my current skills.                                                                                                                 | .28     |
|               | 2           | I use AI only when I am sure I have fully utilized my own abilities and genuinely need its assistance.                                                                              | .51     |
|               | 3           | I focus on enhancing my own skills and use AI solely as a secondary assistive tool.                                                                                                 | .95     |
|               | 4           | I use AI for brainstorming but ensure that the final outcome is the result of my own analysis.                                                                                      | .49     |
| Overreliant   | 5           | Without the assistance of AI-based writing and research tools, completing academic tasks (e.g., articles, theses, or class projects) would be very difficult and exhausting for me. | .74     |
|               | 6           | I have become so reliant on AI tools that I have lost confidence in my own skills.                                                                                                  | .80     |
|               | 7           | Although I have tried to use AI in a limited and controlled manner, I feel that I have become dependent on it.                                                                      | .72     |
|               | 8           | If AI tools are not available, completing academic tasks (e.g., articles, theses, or class projects) independently would be very stressful and challenging for me.                  | .74     |
|               | 9           | I rely on AI tools for preparing a significant portion of my academic work (e.g., articles, theses, or class projects).                                                             | .69     |
|               | 10          | I obtain most of the main ideas and concepts for my research through AI tools, rather than discovering and developing them on my own.                                               | .62     |
|               | 11          | Losing access to AI-assisted writing tools damages my motivation and confidence in completing academic tasks (e.g., articles, theses, or class projects) independently.             | .65     |
| Irresponsible | 12          | Even if I can do it myself, I prefer to use AI for preparing a significant portion of my academic work (e.g., articles, theses, or class projects).                                 | .74     |
|               | 13          | I carefully analyze and review the content generated by AI before using it. (Reversed Item)                                                                                         | .24     |
|               | 14          | I prefer to keep the extent of my use of AI-assisted tools in my academic work (e.g., articles, theses, or class projects) hidden from others.                                      | .38     |
|               | 15          | If a journal questions my use of AI in a manuscript I have submitted, I would deny it, even if I have actually used AI tools to prepare the manuscript.                             | .35     |
|               | 16          | If an AI system is capable of generating a complete paper with minimal effort on my part, I would definitely ask it to prepare the paper for me.                                    | .50     |
|               | 17          | I usually use the information, quotes, citations, or references provided by AI in my work without verifying their accuracy.                                                         | .40     |

Note. AAUI = Academic AI Utilization Inventory

### 2.2.3. The Youth Psychopathic Traits Inventory–Short Version (YPI-S; van Baardewijk et al., 2010)

The YPI-S is a shortened version of the original YPI, consisting of 18 items that are rated on a scale from 1 ("does not apply at all") to 4 ("applies very well"). Items load onto three subscales of GM (e.g., "It is easy for me to manipulate people"), CU (e.g., "To be nervous and worried is a sign of weakness"), and II (e.g., "I consider myself as a pretty impulsive person"), with higher scores indicating more pronounced psychopathic traits. Participants in this study completed the Persian version of the YPI-S, which was validated in Iran by Ebrahimi et al. (2022). Previous research has demonstrated the YPI-S's robust factor structure, reliability, and validity across various adult populations in Iran (Elhami Athar, 2024).

### 2.2.4. The Revised Adult Attachment Scale (RAAS; Collins, 1996)

The RAAS is an 18-item self-report instrument designed to assess adult attachment styles. Respondents rate each item on a 5-point Likert scale, ranging from 1 ("not at all characteristic of me") to 5 ("very characteristic"). The measure evaluates three dimensions of attachment, each represented by six items: Closeness (e.g., "I find it relatively easy to get close to people"), Dependence (e.g., "I am comfortable depending on others"), and Anxiety (e.g., "I find that others are reluctant to get as close as I would like"). The Closeness dimension reflects an individual's comfort with intimacy and emotional connection in relationships. The Dependence dimension assesses the extent to which individuals feel they can rely on and trust others to be available when needed. Finally, the Anxiety dimension captures the degree of worry about being rejected or feeling unloved by partners. The Persian version of the RAAS demonstrated adequate psychometric properties in terms of reliability and validity (e.g., Heshmati et al., 2022).

### 2.2.5. The Rosenberg Self-Esteem Scale (RSES)

The RSES is a 10-item self-report questionnaire developed by Rosenberg (1965) to assess self-esteem. Participants rate items (e.g., "I feel that I have a number of good qualities") on a 4-point Likert scale, with higher scores indicating higher self-esteem. The Persian version of the RSES demonstrated adequate internal consistency and expected correlations with theoretically related constructs (Shapurian et al., 1987).

### 2.3. Data analyses

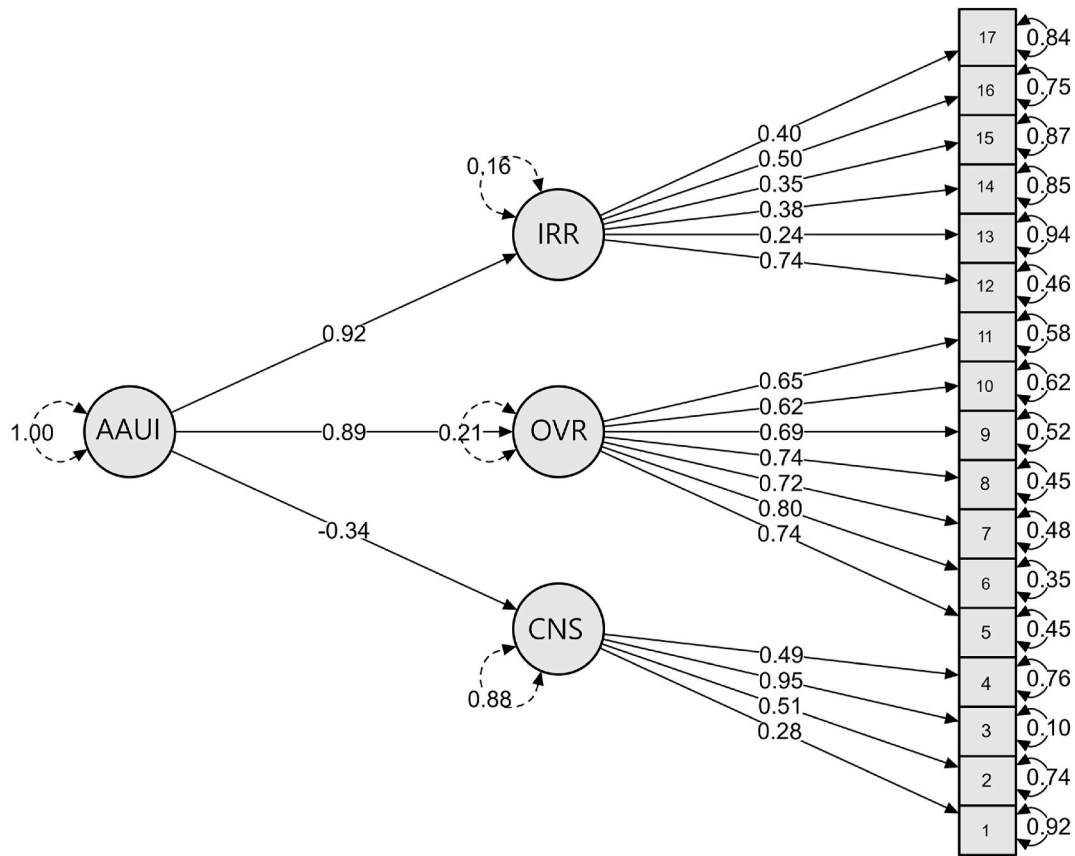
Descriptive statistics were first calculated for all study variables. Pearson product-moment correlation coefficients were then used to explore the relationships among variables, with effect sizes interpreted based on Cohen's (2013) guidelines: small ( $\leq .30$ ), medium ( $.30-.50$ ), and large ( $\geq .50$ ). To maintain focus, only correlations related to the study hypotheses were reported and discussed, with particular emphasis on those of moderate or greater magnitude ( $r \geq .30$ ) as the most meaningful. Subsequently, multiple regression analyses were conducted to test the study hypotheses. Compared to correlation analysis, the regression analyses were considered more robust, as they assess the unique contribution of each predictor to the outcome variables while controlling for the influence of other predictors in the model.

To evaluate the possibility of multicollinearity among predictors before performing multiple regression analysis, variance inflation factors (VIFs) and tolerance values were analyzed. Commonly accepted thresholds suggest that VIFs exceeding 10 and tolerance values below .1 indicate strong correlations that may distort model estimates (e.g., Vittinghoff et al., 2005). In this study, VIFs ranged from 1.23 to 2.12, and tolerance values ranged from .47 to .82, remaining well within acceptable limits. These results indicate that multicollinearity was not a significant concern (full multicollinearity statistics are presented in Supplementary Table S1). To enhance the interpretability of the regression results, squared semi-partial correlations ( $sr^2$ ) were also reported. These values reflect the proportion of unique variance in the outcome variable explained by each predictor, independent of the influence of other variables in the model. Following Cohen's (2013) effect size guidelines,  $sr^2$  values can be interpreted as follows: values below .06 reflect a weak effect, values between .06 and .13 indicate a moderate effect, and values of .14 or higher suggest a strong effect. All analyses were conducted using SPSS version 22.

## 3. Results

### 3.1. Descriptive statistics

Descriptive statistics for all variables, including the mean, standard deviation, skewness, kurtosis, and internal consistency indices, were computed and are reported in Table 2.



**Fig. 2.** Standardized Parameters for the Three-Factor Hierarchical CFA Model of the AAUI

Note. AAUI = Academic AI Utilization Inventory; CNS = Constructive; OVR = Overreliant; IRR = Irresponsible; All factor loadings are significant at  $p < .001$ .

### 3.2. Correlation and regression analyses

#### 3.2.1. Pathological personality traits

As shown in Table 3, Antagonism demonstrated a moderate positive correlation with the AAUI Irresponsible subscale scores ( $r = .33$ ). Similarly, Disinhibition showed moderate positive correlations with both the AAUI Overreliant ( $r = .41$ ) and Irresponsible ( $r = .39$ ) subscale scores. Regression analyses presented in Table 4 indicate that the model comprising six pathological personality traits significantly accounted for 9 % of the variance in the AAUI Constructive subscale scores. Within this model, only Anankastia emerged as a significant positive predictor ( $\beta = .33, p < .001$ ). For the AAUI Overreliant subscale, the pathological traits significantly explained 18 % of the variance, with Disinhibition being the only significant positive predictor ( $\beta = .32, p < .001$ ). Regarding the AAUI Irresponsible subscale, the traits significantly explained 20 % of the variance. Antagonism ( $\beta = .20, p = .03$ ) and Disinhibition ( $\beta = .22, p = .03$ ) were significant positive predictors, while Anankastia ( $\beta = -.20, p = .02$ ) was a significant negative predictor (Table 4).

#### 3.2.2. Psychopathic personality traits

Table 3 reveals two significant correlations of moderate or greater magnitude: the YPI-S CU ( $r = .41$ ) and II ( $r = .31$ ) subscales were positively associated with the AAUI Irresponsible subscale. Regression analyses showed that psychopathic traits significantly explained 9 % of the variance in the AAUI Overreliant subscale, with the YPI-S II subscale ( $\beta = .19, p = .04$ ) being the sole significant predictor (Table 4). For the AAUI Irresponsible subscale, psychopathic traits significantly accounted for 18 % of the variance, with the YPI-S CU subscale emerging as the only significant predictor ( $\beta = .36, p < .001$ ).

#### 3.2.3. Attachment styles

None of the three attachment styles yielded significant correlations of moderate magnitude or greater with the AAUI subscales (Table 3). However, in the regression analyses, attachment styles explained 5 % and 9 % of the variance in the AAUI Overreliant and Irresponsible subscales, respectively. In both models, the Anxious attachment style was the only significant predictor, positively predicting the AAUI Overreliant ( $\beta = .23, p = .006$ ) and Irresponsible ( $\beta = .34, p < .001$ ) subscale scores (Table 4).

#### 3.2.4. Self-esteem

Self-esteem scores yielded a weak positive association with the AAUI Constructive ( $r = .16$ ) subscale score but were negatively and significantly associated with the AAUI Overreliant ( $r = -.20$ ), and Irresponsible ( $r = -.16$ ) subscale scores (Table 3). Furthermore, regression analyses showed that self-esteem significantly explained 3 %, 4 %, and 2 % of the variance in the AAUI Constructive ( $\beta = .16, p = .04$ ), Overreliant ( $\beta = -.20, p = .001$ ), and Irresponsible ( $\beta = -.16, p = .04$ ) subscale scores, respectively (Table 4).

### 4. Discussion

This study examined the relationships between maladaptive personality traits, psychopathic traits, attachment styles, self-esteem, and three distinct academic AI usage profiles: Constructive, Overreliant, and Irresponsible. The results are discussed in detail below.

Regarding maladaptive personality traits, both Antagonism and Disinhibition demonstrated moderate positive correlations with the AAUI Irresponsible subscale scores, while Disinhibition also showed a moderate positive correlation with the AAUI Overreliant subscale scores. Regression analyses further revealed that Disinhibition

**Table 2**  
Descriptive information for study variables (n = 171).

| Measures      | M (SD) | Skewness | Kurtosis | $\omega$ | $\alpha$ | MIC |
|---------------|--------|----------|----------|----------|----------|-----|
| AAUI          | 16.42  | −1.02    | 1.20     | .68      | .64      | .31 |
| Constructive  | (2.87) |          |          |          |          |     |
| AAUI          | 16.95  | .33      | −.81     | .87      | .87      | .50 |
| Overreliant   | (6.86) |          |          |          |          |     |
| AAUI          | 15.38  | .07      | −.52     | .58      | .61      | .20 |
| Irresponsible | (4.42) |          |          |          |          |     |
| Negative      | 7.69   | −.09     | −.69     | .71      | .68      | .25 |
| Affectivity   | (3.83) |          |          |          |          |     |
| Detachment    | 7.36   | .39      | −.23     | .68      | .67      | .25 |
|               | (3.47) |          |          |          |          |     |
| Antagonism    | 6.13   | .59      | .01      | .75      | .75      | .32 |
|               | (3.83) |          |          |          |          |     |
| Disinhibition | 5.45   | .49      | −.20     | .76      | .75      | .34 |
|               | (3.69) |          |          |          |          |     |
| Anankastia    | 10.03  | −.15     | −.62     | .83      | .83      | .45 |
|               | (4.33) |          |          |          |          |     |
| Psychoticism  | 6.68   | .14      | −.62     | .79      | .79      | .39 |
|               | (4.18) |          |          |          |          |     |
| YPI-S GM      | 12.38  | .66      | −.03     | .78      | .78      | .34 |
|               | (4.01) |          |          |          |          |     |
| YPI-S CU      | 10.25  | 1.14     | 1.35     | .70      | .70      | .28 |
|               | (3.32) |          |          |          |          |     |
| YPI-S II      | 10.02  | 1.12     | 1.26     | .72      | .71      | .31 |
|               | (3.06) |          |          |          |          |     |
| AAS Close     | 18.59  | .04      | −.33     | .55      | .60      | .20 |
|               | (4.39) |          |          |          |          |     |
| AAS Depend    | 17.31  | −.06     | .34      | .60      | .62      | .21 |
|               | (4.19) |          |          |          |          |     |
| AAS Anxiety   | 14.22  | .61      | −.18     | .84      | .83      | .45 |
|               | (5.59) |          |          |          |          |     |
| RSES          | 20.92  | −.24     | −.52     | .88      | .88      | .42 |
|               | (5.70) |          |          |          |          |     |

Note. AAUI = Academic AI Utilization Inventory; YPI-S = Youth Psychopathic Traits Inventory-Short Version; GM = Grandiose-Manipulative; CU = Callous-Unemotional; II = Impulsive-Irresponsible; AAS = Adult Attachment Scale; RSES = Rosenberg Self-Esteem Scale;  $\omega$  = McDonald's omega;  $\alpha$  = Cronbach's alpha coefficient; MIC = mean inter-item correlation

significantly accounted for variance in the AAUI Overreliant subscale scores. Similarly, both Disinhibition and Antagonism were significant positive contributors to the variance in the AAUI Irresponsible subscale scores. Conversely, Anankastia was positively associated with the AAUI Constructive subscale scores and negatively associated with the Irresponsible subscale scores. These findings align with theoretical expectations and previous research. Disinhibition, marked by impulsivity and difficulty with self-regulation, could play a significant role in Overreliant and Irresponsible academic AI usage. Impulsive behaviors may lead to excessive or unmonitored dependence on AI tools, while challenges in maintaining focus and exerting deliberate effort could hinder constructive use of such technologies. In line with these, Greitemeyer and Kastenmüller (2023) found a negative relationship between low conscientiousness and the tendency to misuse AI for academic dishonesty. Likewise, Rodríguez-Ruiz, Espejo Siles, and Marín-López (2024) found that low conscientiousness was linked to generating deceptive or fraudulent AI content.

Furthermore, Antagonism, characterized by manipulateness and deceitfulness, could also play a significant role in Irresponsible academic AI usage. Those high in Antagonism may exploit AI tools for unethical purposes, such as plagiarism or hiding their reliance on AI, prioritizing personal gain over academic integrity. This aligns with findings by Greitemeyer and Kastenmüller (2023), who reported positive associations between Dark Triad traits—narcissism, Machiavellianism, and psychopathy—and intentions to use chatbot-generated texts for academic misconduct. Similarly, de Winter et al. (2024) found a moderate correlation between Machiavellianism and an opportunistic approach to concealing the use of ChatGPT in academic work. In addition, the observed relationship between Anankastia and academic AI usage profiles aligns with theoretical expectations. Anankastia, characterized by

**Table 3**  
Correlations between study variables (n = 171).

| Variables                        | 1      | 2      | 3     | 4      | 5      | 6     | 7      | 8      | 9      | 10    | 11    | 12     | 13     | 14     | 15     |
|----------------------------------|--------|--------|-------|--------|--------|-------|--------|--------|--------|-------|-------|--------|--------|--------|--------|
| 1 Constructive                   |        |        |       |        |        |       |        |        |        |       |       |        |        |        |        |
| 2 Overreliant                    | −.23** |        |       |        |        |       |        |        |        |       |       |        |        |        |        |
| 3 Irresponsible                  | −.23** | .58**  |       |        |        |       |        |        |        |       |       |        |        |        |        |
| 4 Negative Affectivity           | −.01   | .23**  | .24** |        |        |       |        |        |        |       |       |        |        |        |        |
| 5 Detachment                     | .00    | .20**  | .23** | .37**  |        |       |        |        |        |       |       |        |        |        |        |
| 6 Antagonism                     | −.04   | .29**  | .33** | .34**  | .33**  |       |        |        |        |       |       |        |        |        |        |
| 7 Disinhibition                  | −.10   | .41**  | .39** | .56**  | .42**  | .55** |        |        |        |       |       |        |        |        |        |
| 8 Anankastia                     | .23**  | .05    | −.03  | .45**  | .32**  | .27** | .19*   |        |        |       |       |        |        |        |        |
| 9 Psychoticism                   | −.04   | .29**  | .21** | .51**  | .47**  | .53** | .54**  | .48**  |        |       |       |        |        |        |        |
| 10 YPI-S Grandiose-Manipulative  | −.06   | .18*   | .19*  | .20**  | .14    | .72** | .36**  | .18*   | .47**  |       |       |        |        |        |        |
| 11 YPI-S Callous-Unemotional     | −.07   | .26**  | .41** | .15    | .34**  | .41** | .27**  | .12    | .30**  | .48** |       |        |        |        |        |
| 12 YPI-S Impulsive-Irresponsible | −.15*  | .27**  | .31** | .41**  | .28**  | .53** | .61**  | .10    | .45**  | .52** | .53** |        |        |        |        |
| 13 AAS Close                     | .02    | −.04   | −.03  | −.26** | −.44** | −.04  | −.10   | −.28** | −.26** | .09   | −.05  | −.08   |        |        |        |
| 14 AAS Depend                    | −.09   | −.06   | −.02  | −.37** | −.50** | −.13  | −.17*  | −.43** | −.41** | −.13  | −.08  | −.17*  | .42**  |        |        |
| 15 AAS Anxiety                   | −.06   | .22**  | .28** | .56**  | .39**  | .33** | .44**  | .26**  | .46**  | .25** | .27** | .50**  | −.31** | −.40** |        |
| 16 Rosenberg Self-Esteem Scale   | .16*   | −.20** | −.16* | −.48** | −.25** | −.11  | −.43** | −.19*  | −.28** | .04   | −.07  | −.34** | .29**  | .16*   | −.34** |

Note. AAUI = Academic AI Utilization Inventory; YPI-S = Youth Psychopathic Traits Inventory-Short Version; AAS = Adult Attachment Scale; \*\* $p < .001$ , \* $p < .05$ .

**Table 4**  
Multiple regression analyses predicting AAUI scores from personality variables (n = 171).

| Outcome Variables |                      |              |                |      |       |       |                 |             |        |                |      |       |       |                 |            |         |                |      |       |       |                 |         |            |
|-------------------|----------------------|--------------|----------------|------|-------|-------|-----------------|-------------|--------|----------------|------|-------|-------|-----------------|------------|---------|----------------|------|-------|-------|-----------------|---------|------------|
| Model             | Predictors           | Constructive |                |      |       |       |                 | Overreliant |        |                |      |       |       | Irresponsible   |            |         |                |      |       |       |                 |         |            |
|                   |                      | F            | R <sup>2</sup> | β    | t     | p     | sr <sup>2</sup> | 95 % CI     | F      | R <sup>2</sup> | β    | t     | p     | sr <sup>2</sup> | 95 % CI    | F       | R <sup>2</sup> | β    | t     | p     | sr <sup>2</sup> | 95 % CI |            |
| 1                 | Negative Affectivity | 2.63*        | .09            | -.07 | -.65  | .52   | .00             | -.20; .10   | 6.12** | .18            | .01  | .15   | .88   | .00             | -.31; .36  | 7.01**  | .20            | .12  | 1.27  | .21   | .01             | .01     | -.08; .35  |
|                   | Detachment           |              |                | .00  | .05   | .96   | .00             | -.14; .15   |        |                | .02  | .26   | .80   | .00             | -.28; .37  |         |                | .10  | 1.28  | .20   | .01             | .01     | -.07; .34  |
|                   | Antagonism           |              |                | -.01 | -.15  | .88   | .00             | -.15; .13   |        |                | .07  | .75   | .45   | .00             | -.20; .44  |         |                | .20  | 2.20  | .03   | .02             | .02     | .02; .43   |
|                   | Disinhibition        |              |                | -.05 | -.47  | .64   | .00             | -.21; .13   |        |                | .32  | 3.10  | <.001 | .05             | .21; .97   |         |                | .22  | 2.20  | .03   | .02             | .02     | .03; .51   |
|                   | Anankastia           |              |                | .33  | 3.64  | <.001 | .07             | .10; .34    |        |                | -.09 | -1.06 | .29   | .01             | -.42; .13  |         |                | -.20 | -2.36 | .02   | .03             | .03     | -.38; -.03 |
|                   | Psychoticism         |              |                | -.13 | -1.22 | .23   | .01             | -.23; .06   |        |                | .10  | 1.02  | .31   | .01             | -.16; .50  |         |                | -.03 | -.26  | .79   | .00             | .00     | -.24; .18  |
| 2                 | YPI-S GM             | 1.37         | .02            | .03  | .27   | .79   | .00             | -.11; .15   | 5.79** | .09            | .01  | .11   | .91   | .00             | -.29; .32  | 12.38** | .18            | -.06 | -.65  | .52   | .00             | .00     | -.25; .12  |
|                   | YPI-S CU             |              |                | .01  | .13   | .90   | .00             | -.15; .17   |        |                | .16  | 1.75  | .08   | .02             | -.04; .70  |         |                | .36  | 4.17  | <.001 | .09             | .09     | .25; .71   |
|                   | YPI-S II             |              |                | -.17 | -1.78 | .08   | .02             | -.34; .02   |        |                | .19  | 1.99  | .04   | .02             | .00; .83   |         |                | .15  | 1.65  | .10   | .01             | .01     | -.04; .46  |
| 3                 | AAS Close            | 1.25         | .02            | .05  | .59   | .56   | .00             | -.08; .14   | 2.80*  | .05            | .02  | .25   | .80   | .00             | -.23; .29  | 5.73**  | .09            | .03  | .39   | .70   | .01             | .01     | -.13; .20  |
|                   | AAS Depend           |              |                | -.16 | -1.75 | .08   | .02             | -.23; .01   |        |                | .03  | .33   | .74   | .00             | -.24; .33  |         |                | .11  | 1.23  | .22   | .01             | .01     | -.07; .29  |
|                   | AAS Anxiety          |              |                | -.11 | -1.31 | .19   | .01             | -.14; .03   |        |                | .23  | 2.79  | .006  | .04             | .08; .49   |         |                | .34  | 4.13  | <.001 | .09             | .09     | .14; .39   |
| 4                 | RSES                 | 4.37**       | .03            | .16  | 2.09  | .04   | .03             | .00; .16    | 7.25** | .04            | -.20 | -2.70 | .001  | .04             | -.42; -.07 | 4.23*   | .02            | -.16 | -2.06 | .04   | .02             | .02     | -.24; .00  |

*Note.* AAUI = Academic AI Utilization Inventory; YPI-S = Youth Psychopathic Traits Inventory-Short Version; GM = Grandiose-Manipulative; CU = Callous-Unemotional; II = Impulsive-Irresponsible; AAS = Adult

Note. AAUI = Academic AI Utilization Inventory; YPI-S = Youth Psychopathic Traits Inventory-Short Version; GM = Grandiose-Manipulative; CU = Callous-Unemotional; II = Impulsive-Irresponsible; AAS = Adult Attachment Scale; RSES = Rosenberg Self-Esteem Scale; sr<sup>2</sup> = Squared Semi-Partial Correlations; \*\*p < .001; \*p < .05.

perfectionism and rigidity, likely fosters Constructive usage by promoting a disciplined and methodical approach to using AI as a supportive academic tool. Simultaneously, its adherence to rules and order may inhibit behaviors associated with Irresponsible usage, such as unethical or careless applications of AI. This highlights how the core traits of Anankastia shape both productive and ethical engagement with AI tools. Meanwhile, given the novelty of these findings, further research is needed to replicate and extend them.

With respect to the psychopathic personality traits, findings showed that the YPI-S CU and II subscales were positively associated with the AAUI Irresponsible subscale, though only the YPI-S CU subscale emerged as the significant correlate in the regression analyses. Also, only the YPI-S II subscale significantly explained the AAUI Overreliant subscale score. The observed relationships between psychopathic personality traits and academic AI usage profiles are consistent with theoretical frameworks and prior research. The CU traits, characterized by a lack of empathy, guilt, and disregard for rules, likely contribute to Irresponsible AI usage, as individuals high in CU traits may engage in unethical behaviors such as plagiarism or undisclosed AI reliance without concern for its potential consequences. On the other hand, II traits, reflecting impulsivity, risk-taking, poor planning, and irresponsibility were significant in explaining both Irresponsible and Overreliant profiles. These traits are theoretically linked to unregulated and excessive behaviors, which can manifest as overreliance on AI tools due to a lack of foresight or self-control. Such tendencies may also hinder deliberate, constructive engagement with AI. These findings are in line with prior research, such as [Greitemeyer and Kastenmüller \(2023\)](#), who found a significant link between psychopathic traits and AI misuse for academic cheating. However, their study predominantly relied on a unidimensional measure of psychopathy, overlooking its multidimensional nature.

Likewise, [Rodríguez-Ruiz, Espejo-Siles, and Marín-López \(2024\)](#) found that psychopathic traits were associated with increased use of AI tools to complete academic work and create fraudulent content. The authors also reported that self-control did not mediate these findings, suggesting that cheating behaviors in higher education for individuals with high psychopathy scores may be planned rather than impulsive. While this study used a multidimensional measure of the psychopathy construct, it is important to note that the authors focused solely on a total psychopathy score in their analyses which contrasts with the prevailing view that studying the individual dimensions of psychopathy is critical for understanding its full impact (for a review see [Elhami Athar, 2024](#)). In this context, findings from the current study underscore the importance of considering the distinct dimensions of psychopathy, as traits such as CU and II exhibited differential relationships with academic AI usage profiles. Therefore, future research should explore these nuanced associations further to deepen our understanding of psychopathy's role in technology-related behaviors, particularly within the context of academia.

Concerning attachment styles, the findings revealed that the Anxious attachment style was the only significant correlate of both the AAUI Overreliant and Irresponsible subscale scores. These results are consistent with theoretical perspectives on attachment and maladaptive coping behaviors. Individuals with an Anxious attachment style tend to exhibit heightened emotional dependency, a fear of failure, and a preoccupation with seeking approval (e.g., [Bartholomew & Horowitz, 1991](#); [Norberg et al., 2018](#)), which may drive their reliance on external tools, such as AI, to compensate for perceived inadequacies. This behavior also aligns with the notion that anxiously attached individuals may turn to non-human sources of support, including technology, as a means to alleviate distress and manage their insecurities ([Mikulincer, 1998](#)). Furthermore, their tendency to experience distress and difficulty managing academic challenges may contribute to Irresponsible usage patterns, such as overusing AI or engaging in unethical practices. This is consistent with evidence suggesting that anxiety heightens threat perception, which can, in turn, lead to self-serving unethical behaviors

(Kouchaki & Desai, 2015). Still, given the novelty of these findings, future empirical studies are necessary to further explore and validate these potential relationships.

Finally, the findings revealed that self-esteem significantly and positively explained variance in the AAUI Constructive subscale scores, while negatively explaining variance in the Overreliant and Irresponsible subscale scores. These results align with evidence that self-esteem influences adaptive and maladaptive behaviors. Higher self-esteem fosters greater confidence and a stronger sense of autonomy, enabling individuals to work independently on academic tasks and use AI as a supplementary tool in a constructive manner. This aligns with previous research showing that higher scores in self-esteem decreased the odds of using AI to address daily doubts due to the need for interacting with someone (e.g., Rodríguez-Ruiz, Marín-López, & Espejo-Siles, 2024). Conversely, lower self-esteem is associated with feelings of inadequacy and external reliance, which could explain its link to Overreliant academic AI usage, as observed in this study. This reliance could stem from a lack of confidence in academic abilities which could foster the overuse of AI in academic contexts (e.g., Ancis, 2020; Tiwari et al., 2023). Furthermore, the observed negative relationship between self-esteem and Irresponsible academic AI usage is in line with the findings of prior research reporting a significant correlation between low self-esteem and high academic dishonesty (e.g., Ancheta et al., 2024). Overall, the findings emphasize the critical role of self-esteem in academic AI behaviors and propose potential directions for future research, especially given the novel insights from this study.

## 5. Limitations and directions for future research

The findings must be considered alongside several limitations. First, the sole reliance on self-report measures introduces shared method variance which could account for some observed relationships between the study variables. Therefore, future research would implement multimethod and multi-informant data collection approaches to address this limitation. Second, the cross-sectional design of this study restricts the ability to establish causal relationships between the variables. Therefore, future research employing longitudinal designs is necessary to examine potential causal links among the study variables. Third, the sample of this study consisted solely of university students, which restricts the generalizability of the results to other academic populations, such as faculty members. Future studies should aim to include a broader range of academic populations, such as faculty members and professionals, to enhance the generalizability of the findings and provide a more comprehensive understanding of academic AI usage across different career stages. Finally, while the sample size of this study was sufficient based on established guidelines for factor analysis and regression, future research should replicate these findings with larger samples to further strengthen the robustness and generalizability of the results.

## 6. Implications for academic integrity

The preliminary findings from this study carry broader implications for how academic institutions address the integration of AI tools in education. First, the emergence of distinct academic AI usage profiles—Constructive, Overreliant, and Irresponsible—underscores the complex nature of students' engagement with technology. These profiles suggest that AI usage is not merely a matter of access or utility but is deeply influenced by individual differences, including personality traits, attachment styles, and self-esteem. This highlights the need for academia to adopt a nuanced perspective that accounts for the psychological factors shaping how students interact with AI in their academic work. Second, the association between maladaptive personality traits, such as Disinhibition and Antagonism, and Irresponsible AI usage raises questions about the ethical challenges posed by emerging technologies. Academic institutions must recognize that while AI tools have

the potential to enhance learning, they also create opportunities for misuse. Understanding these risks can encourage educators to critically assess how AI tools are introduced and monitored within academic environments. Moreover, the links between self-esteem and AI usage suggest that students' psychological well-being plays a central role in shaping their interactions with AI. As higher self-esteem was associated with constructive usage, fostering a supportive and inclusive academic culture that promotes students' confidence may lead to more deliberate and responsible use of AI. Conversely, the relationship between low self-esteem and overreliant or irresponsible usage points to the need for greater awareness of how academic pressures and personal insecurities may drive reliance on AI in ways that undermine academic integrity.

Finally, this study illustrates how technological innovation reshapes educational practices and the broader academic landscape. The findings emphasize the importance of equipping students not only with technical skills to use AI but also with the critical thinking and ethical reasoning necessary to navigate its challenges. By fostering these competencies, academia can ensure that AI serves as a tool for empowerment rather than a source of dependency or ethical compromise.

## 7. Implications for theory

While preliminary, findings from this study offer theoretical implications by demonstrating that well-established links between psychological traits and behavioral patterns in offline contexts are largely mirrored in digital academic settings involving AI use. This continuity reinforces their ecological validity and extends their applicability to contemporary academic behaviors shaped by emerging technologies. For example, Antagonism, typically associated with manipulation, deceit, and disregard for fairness in offline interpersonal interactions, was linked to Irresponsible AI use, suggesting that individuals high in this trait may similarly exploit digital tools in ethically questionable ways. Disinhibition, often characterized by impulsivity, poor planning, and difficulty delaying gratification, was significantly related to both Overreliant and Irresponsible AI usage, reflecting a broader tendency toward unregulated or excessive engagement with academic technologies. In contrast, Anankastia, defined by perfectionism, orderliness, and cautious decision-making, was associated with Constructive AI use and negatively related to misuse, aligning with its offline role in promoting disciplined and ethical behavior across academic tasks.

Similarly, the nuanced pattern of relationships between psychopathy components and academic AI usage profiles aligns with prior research in both offline and digital domains, which suggests that psychopathy components exhibit distinct associations with external correlates (e.g., Elhami Athar, 2025). Moreover, the results contribute to the literature on how interpersonal vulnerabilities manifest in technology use. Anxious attachment, commonly linked to emotional dependency, fear of failure, and heightened sensitivity to approval, was associated with both Overreliant and Irresponsible AI use. This suggests that attachment-related insecurities may translate into maladaptive patterns of digital academic behavior. Similarly, in line with theories that position self-esteem as a protective factor, lower self-esteem was linked to more problematic academic AI usage, whereas higher self-esteem predicted more autonomous and Constructive engagement. These findings extend the significant role of self-esteem and attachment processes into the domain of academic technology use, highlighting their importance in various patterns of digital academic behaviors.

## 8. Conclusion

This study developed and validated the AAUI, a measure designed to assess various academic AI usage profiles, including Constructive, Overreliant, and Irresponsible usage. The AAUI provides a valuable framework for understanding how psychological variables relate to academic AI usage patterns. Specifically, the findings from this study underscore the nuanced ways in which maladaptive personality traits such

as Disinhibition and Antagonism, as well as psychopathic traits like CU and II, contribute to academic AI misuse. Similarly, attachment styles—particularly Anxious attachment—emerged as a significant correlate of Overreliant and Irresponsible AI usage. Self-esteem also played a critical role, with higher self-esteem correlating with more constructive AI use, while lower self-esteem was associated with greater overreliance and misuse. These results offer important insights into the psychological factors influencing academic AI behaviors and highlight the need for continued research in this area. Meanwhile, given the novelty of these findings, future research is essential to enhance the applicability and generalizability of the findings, while advancing our understanding of the complex dynamics behind academic AI usage.

### Ethics approval and consent to participate

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. All Participants provided informed consent after being provided with the study purpose and Assurance of confidentiality.

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### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbr.2025.100679>.

### Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

### References

- Alshater, M. (2022). Exploring the role of artificial intelligence in enhancing academic performance: A case study of ChatGPT. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4312358>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Pub.
- Ancheta, G. J., Fulgencio, L., Santor, K. E., Capagalan, S., Enriquez, C., Santiago, A., et al. (2024). Self-esteem, self-efficacy, and academic dishonesty among freshmen college students: A mixed method study. *Psychology and Education: A Multidisciplinary Journal*, 22, 61–81. <https://doi.org/10.5281/zenodo.12739449>
- Ancis, J. R. (2020). The age of cyberpsychology: An overview. *Technology, Mind, and Behavior*, 1(1). <https://doi.org/10.1037/tmb0000009>. No Pagination Specified.
- Bach, B., Kerber, A., Aluja, A., Bastiaens, T., Keeley, J. W., Claes, L., et al. (2020). International assessment of DSM-5 and ICD-11 personality disorder traits: Toward a common nosology in DSM-5.1. *Psychopathology*, 53(3–4), 179–188. <https://doi.org/10.1159/000507589>
- Bartholomew, K., & Horowitz, L. M. (1991). Attachment styles among young adults: A test of a four-category model. *Journal of Personality and Social Psychology*, 61(2), 226–244. <https://doi.org/10.1037/0022-3514.61.2.226>
- Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, 107(2), 238–246. <https://doi.org/10.1037/0033-2909.107.2.238>
- Bin-Nashwan, S. A., Sadallah, M., & Bouteraa, M. (2023). Use of ChatGPT in academia: Academic integrity hangs in the balance. *Technology in Society*, 75, Article 102370. <https://doi.org/10.1016/j.techsoc.2023.102370>
- Brown, T. A. (2015). *Confirmatory factor analysis for applied research*. Guilford publications.
- Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*. Routledge.
- Collins, N. L. (1996). Revised adult attachment scale. *Behavior Therapy*.
- Cotton, D. R. E., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education & Teaching International*, 61(2), 228–239. <https://doi.org/10.1080/14703297.2023.2190148>
- de Winter, J., Dodou, D., & Eisma, Y. B. (2024). Personality and acceptance as predictors of ChatGPT use. *Discover Psychology*, 4(1), 57. <https://doi.org/10.1007/s44202-024-00161-2>
- Downing, S. M., & Haladyna, T. M. (2006). *Handbook of test development*. Lawrence Erlbaum Associates Publishers.
- Ebrahimi, A., Elhami Athar, M., Bakhshizadeh, M., Fathali Lavasani, F., & Andershed, H. (2022). The Persian version of the Youth psychopathic traits inventory-short version (YPI-S): A psychometric evaluation. *Bulletin of the Menninger Clinic*, 86(1), 48–66. <https://doi.org/10.1521/bumc.2022.86.1.48>
- Elhami Athar, M. (2024). Understanding the relationship between psychopathic traits and client variables: Variable-centered and person-centered analytic approaches. *Current Psychology*, 43(14), 12477–12494. <https://doi.org/10.1007/s12144-023-05322-5>
- Elhami Athar, M. (2025). Exploring the multidimensional nature of the psychopathy construct in social media context: Insights from Instagram. *Computers in Human Behavior Reports*, 17, Article 100603. <https://doi.org/10.1016/j.chbr.2025.100603>
- Greitemeyer, T., & Kastenmüller, A. (2023). HEXACO, the Dark Triad, and Chat GPT: Who is willing to commit academic cheating? *Heliyon*, 9(9), Article e19909. <https://doi.org/10.1016/j.heliyon.2023.e19909>
- Hare, R. (2003). *Hare psychopathy checklist-revised*. Multi Health systems (2nd ed.). Toronto: Inc.
- Heshmati, R., Zemestani, M., & Vujanovic, A. (2022). Associations of childhood maltreatment and attachment styles with romantic breakup grief severity: The role of emotional suppression. *Journal of Interpersonal Violence*, 37(13–14), NP11883–NP11904. <https://doi.org/10.1177/0886260521997438>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- JASP Team. (2024). *JASP (version 0.19.3)[computer software]*. In .
- King, M. R., & ChatGPT. (2023). A conversation on artificial intelligence, chatbots, and plagiarism in higher education. *Cellular and Molecular Bioengineering*, 16(1), 1–2. <https://doi.org/10.1007/s12195-022-00754-8>
- Kouchaki, M., & Desai, S. D. (2015). Anxious, threatened, and also unethical: How anxiety makes individuals feel threatened and commit unethical acts. *Journal of Applied Psychology*, 100(2), 360–375. <https://doi.org/10.1037/a0037796>
- Mikulincer, M. (1998). Attachment working models and the sense of trust: An exploration of interaction goals and affect regulation. *Journal of Personality and Social Psychology*, 74(5), 1209–1224. <https://doi.org/10.1037/0022-3514.74.5.1209>
- Muthén, L. K., & Muthén, B. O. (2002). How to use a Monte Carlo study to decide on sample size and determine power. *Structural Equation Modeling: A Multidisciplinary Journal*, 9(4), 599–620. [https://doi.org/10.1207/S15328007SEM0904\\_8](https://doi.org/10.1207/S15328007SEM0904_8)
- Norberg, M. M., Crone, C., Kwok, C., & Grisham, J. R. (2018). Anxious attachment and excessive acquisition: The mediating roles of anthropomorphism and distress intolerance. *J Behav Addict*, 7(1), 171–180. <https://doi.org/10.1556/2006.7.2018.08>
- Quintans-Júnior, L., Gurgel, R., Araújo, A., Correia, D., & Martins-Filho, P. R. (2023). ChatGPT: The new panacea of the academic world. *Revista da Sociedade Brasileira de Medicina Tropical*, 56. <https://doi.org/10.1590/0037-8682-0060-2023>
- Rodríguez-Ruiz, J., Espejo Siles, R., & Marín-López, I. (2024). The use of artificial intelligence tools among university students and its association with personality traits. <https://doi.org/10.36006/09643-1-10>
- Rodríguez-Ruiz, J., Espejo-Siles, R., & Marín-López, I. (2024). Are psychopathy traits related to the use of artificial intelligence tools among university students? The mediating effect of self-control. *Deviant Behavior*, 1–10. <https://doi.org/10.1080/01639625.2024.2388256>
- Rodríguez-Ruiz, J., Marín-López, I., & Espejo-Siles, R. (2024). Is artificial intelligence use related to self-control, self-esteem and self-efficacy among university students? *Education and Information Technologies*. <https://doi.org/10.1007/s10639-024-12906-6>
- Rohrbach, A., Hendricks, L., Burns, K., Darrell, T., & Saenko, K. (2018). Object hallucination in image captioning. <https://doi.org/10.18653/v1/D18-1437>
- Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton University Press. <http://www.jstor.org/stable/j.ctt183bjph>
- Shapurian, R., Hojat, M., & Nayerahmadi, H. (1987). Psychometric characteristics and dimensionality of a Persian version of Rosenberg self-esteem scale. *Perceptual & Motor Skills*, 65(1), 27–34. <https://doi.org/10.2466/pms.1987.65.1.27>
- Shen, Y., Heacock, L., Elias, J., Hentel, K., Reig, B., Shih, G., et al. (2023). ChatGPT and other large language models are double-edged swords. *Radiology*, 307. <https://doi.org/10.1148/radiol.230163>
- Sullivan, M., Kelly, A., & McLaughlan, P. (2023). ChatGPT in higher education: Considerations for academic integrity and student learning. *Journal of Applied Learning & Teaching*, 6. <https://doi.org/10.37074/jalt.2023.6.1.17>
- Tiwari, A., Chugh, A., & Sharma, A. (2023). Uses of artificial intelligence with human-computer interaction in psychology. In S. Bhatia Khan, S. Namasudra, S. Chandna, A. Mashat, & F. Khafa (Eds.), *Innovations in artificial intelligence and human-computer*

- interaction in the digital era (pp. 173–205). Academic Press. <https://doi.org/10.1016/B978-0-323-99891-8.00003-6>.
- van Baardewijk, Y., Andershed, H., Stegge, H., Nilsson, K. W., Scholte, E., & Vermeiren, R. (2010). Development and tests of short versions of the Youth psychopathic traits inventory and the Youth psychopathic traits inventory-child version. *European Journal of Psychological Assessment*, 26(2), 122–128. <https://doi.org/10.1027/1015-5759/a000017>
- Vittinghoff, E., Glidden, D. V., Shiboski, S. C., & McCulloch, C. E. (2005). *Regression methods in biostatistics: Linear, logistic, survival, and repeated measures models*. Springer Publishing Co.
- World Health Organization. (2019). *International statistical classification of diseases and related health problems (11th Revision)* (Vol. 1). World Health Organization.
- Xiao, Y., & Wang, W. (2021). On hallucination and predictive uncertainty in conditional language generation. <https://doi.org/10.18653/v1/2021.eacl-main.236>.
- Zhang, S., Zhao, X., Zhou, T., & Kim, J. H. (2024). Do you have AI dependency? The roles of academic self-efficacy, academic stress, and performance expectations on problematic AI usage behavior. *International Journal of Educational Technology in Higher Education*, 21(1), 34. <https://doi.org/10.1186/s41239-024-00467-0>